

SOEN 220

Data Communication & Networks

Weeks 1 – 2 | Clean Revision Notes

Software Engineering • Kisii University

1. What is Data Communication?

Data communication is the process of exchanging data or information between two or more devices through a transmission medium such as a wire cable or wireless channel. For communication to occur, the devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).

In simple terms: Data Communication = Transfer of data from a source to a destination in a form that the receiver can understand and use.

2. Components of a Data Communication System

A complete data communication system has five basic components:

1. Message

The information (data) to be communicated. It can be text, numbers, pictures, audio, video, or any combination of these.

2. Sender (Source)

The device that sends the data message. It can be a computer, phone, camera, or any other device capable of sending data.

3. Receiver

The device that receives the message. It must be able to accept and interpret the data sent by the source.

4. Transmission Medium

The physical path through which the message travels from sender to receiver. Examples: twisted-pair cable, coaxial cable, fibre-optic cable, or free space (wireless).

5. Protocol

A set of rules that govern data communication. It represents an agreement between the communicating devices. Without a protocol, devices may be connected but cannot understand each other.

3. Data Flow (Transmission Modes)

Communication between two devices can happen in three different ways depending on the direction of data flow:

Simplex

Data flows in **one direction only**. One device can only send and the other can only receive. Example: Traditional keyboard → computer, or a radio broadcast. The receiver cannot send any data back.

Half-Duplex

Each device can both send and receive, but **not at the same time**. When one device is sending, the other can only receive, and vice versa. Example: Walkie-talkies or older police radio systems.

Full-Duplex

Both devices can send and receive data **simultaneously**. This is the most efficient mode. Example: Modern telephone conversations or high-speed internet connections.

4. Networks and Types of Networks

A network is a set of devices (often called nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data.

4.1 Types of Networks by Size / Coverage

PAN – Personal Area Network

Smallest network, usually covers a range of a few metres. Example: connecting your phone to wireless earphones via Bluetooth.

LAN – Local Area Network

Covers a small geographical area such as a single building, office, or campus. High speed and low error rate. Example: the network inside a university computer lab.

MAN – Metropolitan Area Network

Covers a larger area than a LAN, typically a city or a large campus. Example: a network connecting several branches of a bank within Nairobi.

WAN – Wide Area Network

Covers a very large geographical area (country, continent, or the whole world). The Internet is the largest and best-known WAN.

4.2 Other Important Network Concepts

Internet – A global system of interconnected computer networks that use the Internet Protocol suite (TCP/IP).

Intranet – A private network that uses Internet technologies but is restricted to an organisation's internal users.

Extranet – An intranet that has been extended to allow controlled access by external users (partners, suppliers, customers).

5. Criteria of a Good Data Communication System / Network

When evaluating a network, three main criteria are used:

1. Performance

Depends on the number of users, type of transmission medium, hardware capabilities, and software efficiency. It is often measured by **throughput** (how much data is actually delivered) and **delay / latency** (how long it takes for data to travel from source to destination).

2. Reliability

Measured by how often the network fails, how long it takes to recover from failure, and how robust it is against disasters or attacks. A reliable network has high availability and good fault tolerance.

3. Security

Protecting data from unauthorised access, damage, or modification. This includes physical security of equipment as well as protection against viruses, eavesdropping, and cyber attacks.

6. Network Topology

Topology refers to the **layout** or arrangement of devices and connections in a network. There are two main views of topology:

- **Physical Topology** – The actual physical arrangement of cables and devices (what you can see and touch).
- **Logical Topology** – How data actually flows in the network (which can be different from the physical layout).

Common Physical Topologies:

Mesh

Every device is connected to every other device. Very reliable but expensive and complex. Used in critical systems.

Star

All devices connect to a central hub or switch. Easy to install and manage. If the centre fails, the whole network goes down.

Bus

All devices share a single backbone cable. Simple and cheap, but if the main cable fails, the entire network fails.

Ring

Each device connects to exactly two other devices, forming a circle. Data travels in one direction. Failure of one device can affect the whole ring unless dual rings are used.

Hybrid

Combination of two or more topologies (e.g., star-bus or star-ring). Most large modern networks are hybrid.

7. Transmission Media

The transmission medium is the physical path between transmitter and receiver. It is broadly divided into two categories:

7.1 Guided Media (Wired)

Signals are guided along a solid medium. Common types:

- **Twisted-Pair Cable** – Two insulated copper wires twisted together. Cheap and widely used in LANs (e.g., Ethernet cables).
- **Coaxial Cable** – Central conductor surrounded by insulating layer, metallic shield, and outer plastic cover. Better shielding than twisted pair.
- **Fibre-Optic Cable** – Uses light pulses through glass or plastic fibres. Extremely high bandwidth, immune to electromagnetic interference, and suitable for long distances.

7.2 Unguided Media (Wireless)

Signals travel through air or free space without a physical conductor. Examples include radio waves, microwaves, infrared, and satellite communication. Wireless is more flexible but generally more susceptible to interference and security issues.

8. Switching Techniques

When data needs to travel across a network with multiple intermediate nodes, switching decides how the data is forwarded. Three main switching techniques exist:

Circuit Switching

A dedicated communication path is established between sender and receiver for the entire duration of the conversation. Used in traditional telephone networks. Three phases: **Path establishment** → **Data transfer** → **Path termination**. Resources are reserved even if no data is being sent (can be inefficient).

Packet Switching

Data is broken into small packets. Each packet is sent independently and may take different routes. Packets are reassembled at the destination. More efficient use of network resources. This is how the Internet works.

Message Switching

The entire message is sent from node to node (store-and-forward). Each intermediate node receives the full message, stores it, and then forwards it. Less common today compared to packet switching.

9. Types of Connections

- **Point-to-Point** – Direct dedicated link between two devices. Simple and secure but does not scale well.
- **Multipoint / Multidrop** – More than two devices share a single link. Cheaper but requires mechanisms to control who transmits when.
- **Broadcast** – One device sends and all other devices on the network receive the message (classic example: traditional Ethernet hub).

10. Data Transmission Process (Simplified)

A typical flow when sending data looks like this:

1. **Source** generates the raw data (message).
2. Data is **encoded** into a form suitable for transmission (bits, signals).
3. The encoded signal is **transmitted** over the medium.
4. At the receiving end the signal is **decoded** back into usable data.
5. The **receiver** gets the original message.

Protocols and error-control mechanisms work throughout this process to ensure the data arrives correctly and in order.

11. Important Supporting Concepts

Flow Control

A technique that ensures the sender does not overwhelm the receiver with data faster than it can process. It prevents buffer overflow and data loss.

Error Detection & Correction

Methods used to discover whether data has been corrupted during transmission and, if possible, to correct it. Common techniques include parity bits, checksums, and CRC (Cyclic Redundancy Check).

Addressing

Every device on a network needs a unique address so that data can be delivered to the correct destination. Examples: MAC addresses (physical) and IP addresses (logical).

Routing

The process of selecting the best path for data to travel from source to destination across multiple networks. Routers perform this function using routing tables and protocols.

Network Management

Activities required to keep a network running efficiently and securely. Key areas include:

- Security management
- Performance / bandwidth monitoring
- Fault detection and recovery
- Configuration management
- Accounting (who is using what resources)

12. Advantages of Networking

- Resource sharing (printers, files, internet connection)
- Better communication and collaboration
- Centralised data management and backup
- Cost reduction through shared hardware and software
- Improved reliability and redundancy (if designed well)
- Easier expansion and scalability

13. Reference Models (Brief Introduction)

To standardise how different systems communicate, two important models were developed:

- **OSI Model (Open Systems Interconnection)** – A 7-layer theoretical model developed by ISO. It is excellent for understanding and teaching networking concepts.
- **TCP/IP Model** – The practical model actually used on the Internet. It has 4 (or 5) layers and is the foundation of modern networking.

You will study both models in more detail in later weeks. For now, just remember that they provide a structured way to think about the different functions needed for successful data communication.

14. Emerging Issues in Data Communication

Modern networks face several ongoing challenges:

- Ever-increasing demand for higher bandwidth and lower latency
- Security threats (hacking, malware, phishing, DDoS attacks)
- Privacy concerns with large-scale data collection
- Network congestion and quality of service (QoS)
- Integration of IoT (Internet of Things) devices at massive scale
- Energy efficiency of network infrastructure

These notes cover the foundational concepts from the first two weeks of SOEN 220. Master these ideas well — almost everything later in the unit builds on them.

Tip for revision: Try to explain each concept out loud in your own words. If you can teach it to a classmate, you truly understand it.

SOEN 220 – Network Structure & Topologies

Data Communication & Networks | Clean Revision Notes

1. Intranet and Extranet

Intranet

A private network that is accessed **only by members of a firm / organisation**. It uses the same technologies as the Internet (TCP/IP, web browsers, etc.) but is restricted to internal users. Typical uses: company policies, internal email, shared files, HR systems.

Extranet

A private network that is accessed by members of the organisation **and by trusted external partners** (suppliers, customers, contractors). It extends the intranet outward in a controlled way. Access is usually protected by passwords, VPNs or firewalls.

Simple memory aid: **Intra** = inside only. **Extra** = inside + selected outsiders.

2. Network Topology

Topology

The virtual shape or structure of a network — how devices are arranged and connected.

Physical topology

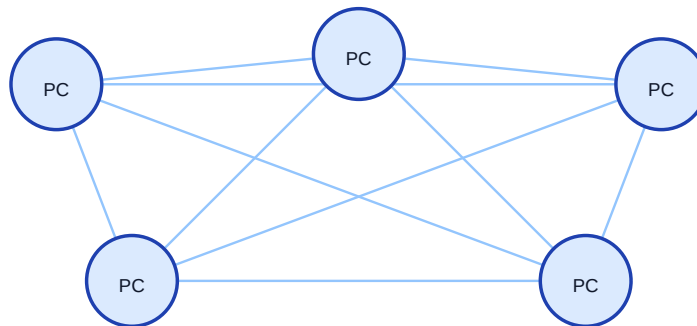
The actual way in which a network is laid out physically (the real cables and device positions).

When designing a network, three important factors are always considered:

- **Performance** – how fast and efficiently the network responds and transfers data
- **Reliability** – how correctly and consistently the network works, and how well it recovers from failures
- **Security** – the ability to protect data and resources from unauthorised access or attacks

3. Mesh Topology

In a mesh topology every device has a dedicated point-to-point link to every other device. A dedicated link carries traffic only between the two devices it connects.



Every device linked to every other (full mesh)

Full mesh – each node connected to all others

Advantages

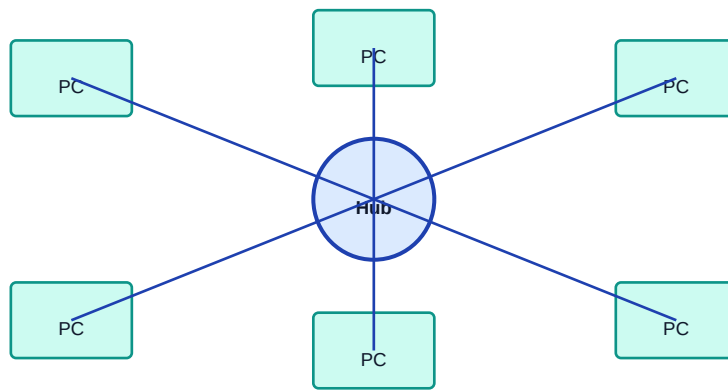
- Use of dedicated links guarantees that each connection carries only its own data (no sharing of bandwidth on that link)
- Robust – a fault in one link does not incapacitate the entire system
- Privacy and security are high because data travels on a private path between two devices

Disadvantages

- Requires a lot of cabling and many I/O ports
- Installation and reconnection are difficult
- Hardware required is expensive

4. Star Topology

Each device has a dedicated point-to-point link only to a central controller (usually a hub or switch). Devices do not connect directly to each other; all traffic goes through the centre.



Star – all devices linked to a central hub/switch

Advantages

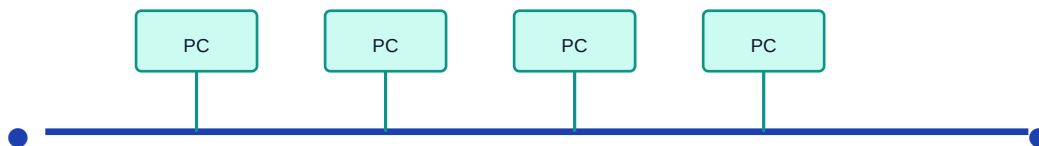
- Less expensive than mesh (each device needs only one link)
- Easy to install and reconfigure – add or remove a device without disturbing the rest
- Robustness – failure of one link or one device does not affect the others
- Easy to identify and isolate faults

Disadvantages

- Dependency on the central hub – if the hub fails, the whole network fails (single point of failure)

5. Bus Topology

All devices share one long cable (the backbone). Signals travel along the cable in both directions. A multipoint configuration: one cable, many stations connected to it.



Shared backbone cable

Bus – one shared backbone cable with stations attached

Advantages

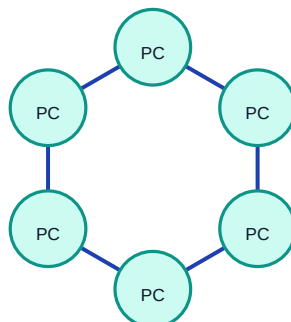
- Easy to install
- Uses less cabling than star or mesh → cheaper for small networks

Disadvantages

- Difficult reconnection and fault isolation
- A break in the bus cable can stop all transmission
- Signal reflection at the ends / taps can cause degradation of quality

6. Ring Topology

Each device has a dedicated point-to-point connection with only the two devices on either side of it. Data travels in one direction around the ring (or sometimes in both directions in dual rings).



Ring – devices form a closed loop

Repeaters are often used to regenerate the signal as it travels around the ring so that it does not become too weak.

Advantages

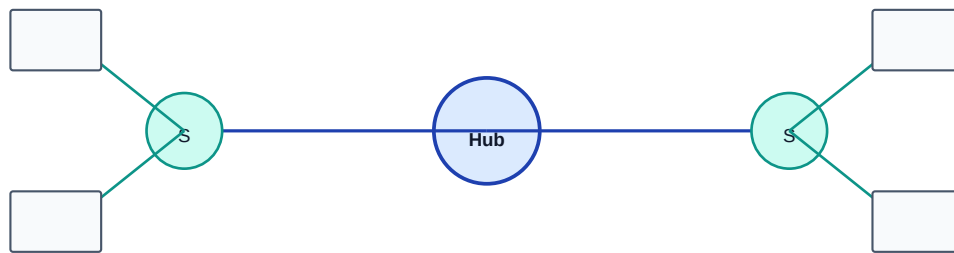
- Easy to install and reconfigure
- Fault isolation can be simplified (a break can be detected by the neighbouring stations)

Disadvantages

- A break in one point can disrupt the whole ring (unless a dual-ring or bypass mechanism is used)

7. Hybrid Topology

A hybrid topology combines two or more different types of topology. A common example is a star-of-stars: several star networks connected together through a central hub or switch.



Hybrid = combination of two or more topologies (e.g. star + star)

Hybrid example – multiple stars joined together

Hybrid designs are used when no single topology is ideal for the whole organisation. Different departments can use the topology that suits them best, then be linked together.

8. Factors to Consider When Choosing a Topology

1. **Nature of the network** – size, purpose, number of users
2. **Cost** – budget for cables, devices and installation
3. **Length of the cable needed** – longer runs cost more and can affect signal quality
4. **Future growth** – will it be easy to add more devices later?
5. **Cable type** – twisted pair, coaxial, fibre, etc., and the distances they support

9. Network and Internet (quick reminder)

A **network** is the connection of computers and devices for the purpose of communication and sharing of resources. The **Internet** is the global interconnection of networks — a network of networks that spans the world.

10. Quick Revision Checklist

- Intranet = private, organisation only. Extranet = organisation + trusted outsiders.
- Topology = the shape/structure of the network (physical or logical).
- Mesh = every device linked to every other → very reliable, very expensive.
- Star = all devices linked to a central hub → easy to manage, hub is a single point of failure.
- Bus = one shared cable → cheap, but a cable break can stop everything.
- Ring = closed loop → simple, but one break can affect the ring.
- Hybrid = mix of topologies → flexible for large organisations.
- Always weigh performance, reliability, security, cost and future growth when choosing.

These notes expand the class material on network topology and structure for SOEN 220. Draw the five topology diagrams yourself a few times — that is the fastest way to lock them in memory before the CAT.